

Impact of buyer-supplier TMT misalignment on environmental performance

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Abstract

Purpose – This study investigates the relationship between buyer-supplier top management team (TMT) demographic misalignment (defined as differences in TMT composition based on background, age and gender) and environmental performance (EVP).

Design/methodology/approach – The empirical setting is publicly held US manufacturing firms that are present in both the Kinder, Lydenberg and Domini's (KLD's) annual EVP ratings and Bloomberg's supply chain database. The study employs panel data regression methods on an unbalanced panel dataset of 7,493 dyad-year observations comprising 427 unique firms.

Findings – The research shows that misalignment in functional background and gender composition between TMTs have a negative outcome on both the buyer's and the suppliers' EVP. However, increasing presence of females across TMTs has a positive influence on EVP. Further, the research shows that misalignment based on age between the TMTs does not impact EVP in any significant way. On the contrary, increasing age across TMTs is a significant predictor of EVP.

Originality/value – This study builds on existing works in TMT heterogeneity and adds context to the heightening belief in the positive linkage between heterogeneity and performance through extension to a boundary spanning interfirm context.

Keywords Environmental management, Buyer-supplier relationships, Top management team, Alignment, Regression analysis

Paper type Research paper

Introduction

When there is alignment and understanding, it is much easier to navigate forward together, moving in and out of agreement. – Karen Kimsey-House, *Co-Active Leadership: Five Ways to Lead*

The alignment of priorities and objectives between a firm and its suppliers is arguably the defining concept of supply chain management, and this alignment is achieved through interorganizational relationships (Selviaridis and Spring, 2018). Walker *et al.* (2014) have highlighted that misalignment in western codes of conduct of multinational corporations and the local culture of suppliers in developing countries become a barrier to implementation of socially sustainable practices. Given the focus on environmental performance (EVP), it has been determined that supplier buy-in or buyer-supplier alignment is instrumental in achieving the environmental goals of a firm (Carter and Carter, 1998; Dwayne Whitten *et al.*, 2012; Vachon and Klassen, 2006; Wilhelm *et al.*, 2016), and consequently, of the entire supply



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chain. In the quest for external alignment, the extant literature has looked at several different buyer-supplier factors including proper incentives for suppliers, contractual arrangements and cultural orientation among others (Cannon *et al.*, 2010; Cao and Lumineau, 2015). Research has proposed that governance misalignment heightens coordination challenges, and ultimately, negatively impacts supply chain outcomes such as the buyer's perception of responsiveness, quality and reliability through heightened coordination challenges (Handley, 2017). An important and yet understudied aspect concerns the nature of alignment or misalignment between buyer-supplier top management teams (TMTs hereafter). The importance of TMTs in formulating EVP strategy has been established (Buysse and Verbeke, 2003; Kumar and Paraskevas, 2018; Russo and Fouts, 1997; Sharma, 2000). Buyer-supplier TMTs have a default faultline driving schisms due to their respective group identification (see Li and Hambrick, 2005). As noted by the Li and Hambrick (2005), demographic similarity (dissimilarity) can reduce (enhance) the effect of natural faultlines. It has been established that demographic similarity between entities leads to greater cooperative interaction, better information exchange and rapport (Simsek *et al.*, 2005), which results in goal congruence and better performance (Golicic and Mentzer, 2005). On the other hand, demographic dissimilarity impairs collaboration, information exchange and joint decision-making (Simsek *et al.*, 2005). Therefore, it is important to examine the forms of demographic misalignment between TMTs as an interorganizational barrier impacting EVP.

Prior research has emphasized the importance of relational demography, defined as "the comparative demographic characteristics of members of dyads or groups" (Tsui and O'reilly, 1989, p. 403). For example, researchers have looked at similarity in age, gender, experiences amongst other characteristics within groups to assess influence on various business outcomes (Jackson *et al.*, 1991; Wiersema and Bantel, 1992; Wiersema and Bird, 1993; Zajac and Westphal, 1995). However, there remains a theoretical gap in the literature regarding the buyer-supplier relational demography between TMTs on EVP. This study investigates the relationship between buyer-supplier demographic TMT misalignment and the EVP of the supply chain, theorizing that misalignment will be an interorganizational barrier impacting EVP. Specifically, the study conceptualizes three forms of demographic misalignment between a buyer and its supplier's TMT, i.e. functional background, age and gender, and theorizes their linkages to supply chain EVP. In addition, the study further investigates the linkages among these conceptual misalignment forms on the buyer's as well as the suppliers' EVP separately, gauging the extent of differing impact on each party: buyer and supplier.

This research fills an important gap as studies have shown the importance of TMTs in setting the environmental and social agenda of the firm (Alexiev *et al.*, 2010; Chen *et al.*, 2010; Hambrick, 2007; Hambrick *et al.*, 1996; Hambrick and Wowak, 2019; Kumar and Paraskevas, 2018; Reimer *et al.*, 2018), however, extant research examining the impact of the interplay between buyer-supplier TMT characteristic as an external factor impacting EVP is very limited. To the best of the authors' knowledge, no study has explored the impact of demographic misalignment between the buyer and its supplier's TMT on EVP. While findings related to TMT heterogeneity suggest the importance of diversity within TMTs (Hambrick *et al.*, 1996), it risks downplaying the importance of TMT misalignment between firms. In this context across groups, it is important to note the importance of shared vision and viewpoints that complement each other for improved performance. This study, therefore, extends theory regarding how misalignment between buyer and supplier TMTs influence EVP by leveraging upper echelons theory (UET) and the complementarity theory.

This study posits that complementarities (or the lack thereof) in EVP strategy between buyer and supplier TMTs is highlighted by the level of demographic alignment (misalignment), which will impact EVP of not only the buyer, but the entire supply chain including the suppliers. In doing so, this study extends the complementarity viewpoint in two significant ways: (1) beyond traditional applications of structural and process governance

(Koza and Dant, 2007), or asset complementarities (Darnall and Edwards, 2006), to complementarities from demographic alignment that would lead to behavioral integration from a relational perspective and (2) by juxtaposing it with UET for strengthening the arguments around the lack of behavioral integration due to misalignment across TMTs and the impact on chain-level EVP. Specifically, the research questions in this study are: Does misalignment of functional background between buyer and suppliers' TMTs affect EVP? Does misalignment of age characteristics between buyer and suppliers' TMTs affect EVP? Does misalignment of gender characteristics between buyer and suppliers' TMTs affect EVP? In addressing these questions, this study builds on the existing body of knowledge in the TMT and environmental management (EM) literatures by examining how interorganizational factors have an impact on EVP.

This study makes a significant theoretical and managerial contribution to the TMT and EM literature studies. This research contributes to theory by juxtaposing UET's central tenet of the importance of upper management in decision-making with the importance of complementarities generated from relational demography. Specifically, this study proposes that key mechanisms of behavioral integration can be applied in relationships between firms as opposed to only within an individual firm. This paper further contributes to UET and complementarity theory by developing unique hypotheses, which propose that misalignment between buyer and supplier TMTs will act as an organizational barrier to maximizing EVP. The theoretical relationships are empirically tested using an original panel dataset of 7,493 dyad-year observations derived from Bloomberg's Supply Chain (SPLC) analysis data, the Bloomberg executive databases, the Kinder, Lydenberg and Domini (KLD) data and Compustat data. The main findings support the consequences of functional and gender misalignment between TMTs on EVP, though it does not find support for demographic misalignment with respect to age. The managerial implications highlight the importance of accounting for demographic dissimilarities in strategic planning sessions.

The paper is organized as follows: first, the theory and hypotheses section is developed. Next, the methodology section where the data are described, variables defined and model specified. The subsequent section highlights the results of the econometric model and robustness checks. Finally, the results, theoretical contribution and managerial implications are discussed with suggestions for future research.

Theory and hypotheses

Upper echelons theory (UET)

UET has its origins in the Behavioral Theory of the Firm (Hambrick and Mason, 1984) and asserts that the decisions a manager makes are based on their interpretations of strategic events and that a manager's interpretation of strategic events is informed by her/his experiences, biases and personality (Hambrick, 2007). UET follows the Behavioral Theory of the Firm assuming that a manager's decisions are constrained by the limitations of bounded rationality and imperfect information, specifically that managers "...are significantly constrained by limitations of information and calculation. Because of those limitations, explicit and timely calculations of optimality are costly or impossible" (Cyert and March, 1963). Hambrick and Mason's seminal article in 1984 introduced UET and proposed that managers first perceive the relevant information in the world around them through a mechanism called the field of vision, which limits the amount of information processed. Subsequently, they reduce the scope of information through a process called selective perception, which focuses their attention to relevant pieces of available information based on their experience, temperament, values, background, priorities, etc. and forms the basis for actionable strategy. Finally, their interpretations of the remaining information and priorities lead to strategic decision-making.

Therefore, a manager's functional background, experiences and values all play a role in this process of turning information into strategic decisions (Hambrick and Mason, 1984).

Research has shown that heterogeneity in functional backgrounds on TMTs impact frequency, noteworthiness and speed of firm actions (Hambrick *et al.*, 1996). Furthermore, it has been shown that firms having a CEO with higher degrees of career variety have increased levels of strategic dynamism (Crossland *et al.*, 2014). TMT age and functional diversity are shown to impact new product performance (Armstrong *et al.*, 2010). Specifically, political liberalism (Chin *et al.*, 2013), CEO education (Lewis *et al.*, 2014), CEO narcissism (Petrenko *et al.*, 2016), TMT interactive complexity and TMT decentralization of decision-making (Wong *et al.*, 2011) and TMT supply chain experience and demographics (Kumar and Paraskevas, 2018) have all been shown to impact the EVP of the firm.

Complementarity theory

Complementarity theory was introduced by Edgeworth (1881) in the context of mathematical physics where activities together resulted in an effect that was more than individual outcomes. This was extended to organizations by Milgrom and Roberts (1995) who defined complementary actions by firms as those that mutually enhanced their respective contributions. The theory advocates complementarity between or across groups/firms from a distinct skills, resources, products and assets perspective with specific use found in alliance and merger studies (for, e.g. see Yu *et al.*, 2016).

In the environmental context, Christmann (2000) finds that complementary assets in the form of process innovation, and implementation moderates the impact of best practices on performance. Darnall and Edwards (2006) find that the adoption of EM systems is eased in the presence of complementary assets. Similarly, Paulraj *et al.* (2014) examine the complementary effects of structural and process governance mechanisms on environmental collaboration between supply partners. Importantly, Cassiman and Veugelers (2006) suggest that the complementarities between two firms' organizational design choices will be sensitive to elements of a firm's strategic environment.

This paper extends the theories of UET and complementarity by considering how a lack of demographic alignment in TMT characteristics impact EVP. UET advocates for the importance of unique experience of TMT members in influencing the strategic environment and shaping the agenda of the firm. Complementarity builds on the unique experience by highlighting the importance of commonalities and the risks from a lack of alignment in buyer-supplier TMT characteristics in the pursuit of strategic goals and objectives.

The role of TMT demographics

For developing a shared strategic vision on the TMT, Preston and Karahanna (2009) examine age, gender and functional background to assess common interests among members. In addition, Kumar and Paraskevas (2018) examine the role of functional background, age and gender characteristics of a TMT and the impact on EVP. This paper extends this stream of research by hypothesizing the impact of misalignment based on the dissimilarity in these three attributes between buyer TMT's and its suppliers on EVP. In doing so, this research emphasizes the importance of commonalities between TMTs in the pursuit of environmental objectives, which takes greater salience given the uncertainties of favorable financial outcomes.

For superior EVP, buy-in must be achieved between supply chain partners for optimal chain-level outcomes. For example, an important step in implementing green sourcing processes is to assess the opportunities throughout the value chain for elimination of waste and environmentally irresponsible production (Chase and Jacobs, 2008). It is therefore important to ascertain if supply chain partners are willing and capable of delivering

collaborative processes and strategies in accordance with the requested EVP objectives (Chase and Jacobs, 2008). For collaborative competence, Whipple *et al.* (2015) emphasize the importance of building external relationships. It has been noted that long term buyer-supplier relationships lead to superior supply chain performance (Dey *et al.*, 2015). Along these lines, Tsanos and Zografos (2016) have established that external relationships are built through mechanisms such as information integration.

UET states that behavioral integration on the TMT impacts firm strategies (Simsek *et al.*, 2005). Behavioral integration of the members on the TMT aids in success of strategic outcomes for the firm (Hambrick, 1994). Specifically behavioral integration has been found to improve organizational ambidexterity (Carmeli and Halevi, 2009), develop global strategies (Hambrick, 1998), reduces cognitive conflict (Li and Zhang, 2002) and is particularly important in rapidly changing environments (Carmeli and Schaubroeck, 2006), whereas behavioral dis(integration) is detrimental to performance (Lubatkin *et al.*, 2006). UET posits that behavioral integration is present in TMTs when they exhibit informational exchange, collaborative behavior and joint decision-making (Hambrick, 1994). Similarities or alignment in functional background across TMTs will result in similar viewpoints and shared interests that will aid in the process of behavioral integration, while misalignment will impede the process of such integration. In this respect, demographic heterogeneity has been examined as a source of relationship conflict within TMTs (Amason *et al.*, 2018). It has been shown that demographic heterogeneity inhibits interpersonal connections and obstructs informal communications (Amason *et al.*, 2018; Smith *et al.*, 1994). In the group context, where each TMT is treated as a distinct group, the study by Li and Hambrick (2005) can be applied to understand the impact of demographic similarities on natural faultlines that exists between two groups. As noted by the authors, behavioral integration from demographic similarity will help bridge the schisms associated with identity related faultlines.

The intersection of the complementarity viewpoint and UET guides us to the understanding that behavioral integration within individual TMTs is not enough to achieve organizational outcomes, which require joint buy-in from a buyer's TMT and its supplier's TMT. Since EVP strategies require buy-in across firms to implement, this paper theorizes that it is critical to have behavioral integration between the TMTs of the firms engaging in strategic environmental initiatives. In line with prior studies assessing TMT demographics (Amason *et al.*, 2018), this research considers alignment (or lack thereof) on the basis of functional background, age and gender as elements of behavioral integration to assess the impact on EVP.

TMT functional background misalignment

Research has established that firm level environmental strategies are influenced by the C-Suite (Kumar and Paraskevas, 2018). For instance, the likelihood of a firm disclosing environmental impacts are, in part, driven by whether the CEO has an MBA or a law degree (Lewis *et al.*, 2014). In this respect, the functional background of TMT members, as an element of TMT demographics, influences the choice of environmental strategy. For example, research indicates that accountants could bring the traditional functions of their discipline to EM processes in terms of measuring, monitoring, recording and verifying financial data (Wilmschurst and Frost, 2001). In contrast, Nath and Mahajan (2008) note that a chief marketing officer (CMO) is primarily concerned with branding and messaging of the firm to increase a firm's attractiveness to stakeholders. These studies highlight the challenges of developing a coherent strategy within a multidisciplinary TMT and the amplification of these challenges in building consensus when priorities differ across firm boundaries.

The level of behavioral integration is constrained by a manager's field of vision and selective perception. For instance, TMTs which are dominated by operations expertise may have a field of vision which primarily encompasses the manufacturing process, quality signals and raw material shortages (Chase and Jacobs, 2008); while teams dominated by marketing experience may have a field of vision which focuses more heavily on the demand signals from customers, customer complaints and the efficacy of advertisements (Armstrong, 2009). It intuitively follows that a team which is dominated by operations experience will have a significantly different subset of the information in their field of vision than a team dominated by marketing experience (Hambrick and Mason, 1984). Furthermore, executives with different backgrounds will also likely have a different interpretation process for information that they have gathered. While an operations management executive may interpret information through the lens of efficiency and reduction of waste, a financial management executive may look at the same information through a lens of systemic risk and expected returns. Therefore, this study proposes that a lack of alignment in functional background will impede consensual processes of strategic decision-making and negatively impact EVP.

TMTs misaligned in terms of functional composition will also have lower levels of behavioral integration due to the complexity of information exchange. Information exchange is a function of the richness, timeliness and accuracy of communications (Hambrick, 1994). TMTs that are aligned through similarities in functional background will be able to communicate with fewer time-consuming misunderstandings, as their common backgrounds will aid them in understanding nuanced situations with common jargon and shared experiences. Joint decision-making will be more challenging in situations with misaligned TMTs because, as stated previously, the process of interpreting information will be different for executives with different backgrounds, likely leading to different suggested strategies for environmental improvement. While extant research does cite the benefit of teams with diverse background in their ability to scout for new sources of information, innovate and adapt; homogeneity across TMT teams increases conformity, group identification, cohesivity (Wiersema and Bantel, 1992) that will increase the buy-in toward improved EVP.

In sum, the idea of homogeneity in functional background across TMTs creating complementary teams is grounded in the extant literature (Krishnan *et al.*, 1997). The cultural fit, communication, cooperation and shared understanding of complementary teams will result in improved performance. Therefore, guided by UET and complementarity, the functional alignment between the TMTs of both the buyer and supplier firm will be beneficial in implementation of environmental initiatives. On the other hand, a functional disconnect can lead to greater wrangling and discord. Furthermore, a lack of consensus and disjointed pursuit of overall environmental objectives will be detrimental to performance. Accordingly, hypothesis one states:

H1. Higher levels of TMT functional misalignment will be negatively related to EVP.

TMT gender alignment

Toward a sustainable agenda, women play a significant role as proponents of environmentally friendly initiatives (Brough *et al.*, 2016). For example, gender diversity on the board has been found to have a positive influence on voluntary disclosures by the firm (Ben-Amar *et al.*, 2017). Women have a distinct viewpoint on environmental issues. In a study on gender, values and environmentalism, Kalof *et al.* (2002) find that underlying altruism is a stronger motivation for women as compared to men. Tindall *et al.* (2003) also find a higher level of environmental activism in women as compared to men. The difference in environmentalism between genders has been partly attributed to empathy and social dominance (Milfont and Sibley 2016; Stanley *et al.*, 2017). According to the authors, empathy

drives women while social dominance forms the driver for men toward environmentalism. This can be extended to suggest that the types of environmentally friendly actions supported by men can be distinctly different from women due to different underlying motivations. For instance, men might be more likely to support visible actions such as, green products, to improve social standing, whereas women are more likely to support actions surrounding cleanups, emission controls, etc. with an altruistic appeal to address the dire predictions from ignoring such issues. It is therefore important to consider the distinct gender attitudes toward environmentalism in looking for synergies between groups with different levels of gender composition. Anecdotal evidence (Paquette, 2016) suggests that women are more inclined to have an environmentally friendly posture. Brough *et al.* (2016) find evidence that green behavior is cognitively linked with femininity. As a gender group, they find that women are more concerned about climate change and tend to use less fuel and energy. The presence of women on buyer and supplier TMTs, then, would help define an environmental agenda that is distinct from the environmental priorities of male members of the TMT.

UET suggests that executives with different backgrounds and values will selectively perceive different information when faced with a similar set of stimuli (Hambrick and Mason, 1984). This paper proposes that the differences in information perception and ultimate strategic choices in the environmental arena between TMTs with varying levels of gender composition will lead to difficulty in the behavioral integration process between two such teams.

Behavioral integration between TMTs with varying levels of gender composition in support of a common environmental agenda can be further impeded by communication and interaction styles that differ by gender. UET research indicates that communication is a critical component of behavioral integration (Hambrick, 1994) and extant research has indicated that gender can influence communication and interaction styles (Callan, 1993). Similar levels of gender presence can be viewed as a relational complementarity between TMTs that helps with communication and overall working relationship between TMTs. Extending the complementarity viewpoint, this study theorizes that gender alignment improves homogeneity and will improve communications, increase solidarity and positively impact the working relationship between buyer and supplier TMTs.

Following UET logic of unique gender perspective based on selective perception, and applying that via the complementarity perspective, this study postulates that greater alignment between buyer TMT and its supplier's TMT in terms of gender composition can achieve higher levels of EVP. Teams that are misaligned in their gender composition will be less likely to share a common agenda or communicate effectively and struggle with a lack of behavioral integration in pursuit of environmental strategies.

Leveraging the above arguments on the impact of respective gender composition on communication, purpose and priorities, this paper formally hypothesizes the relationship between gender misalignment and EVP as follows:

- H2.* Higher levels of TMT gender misalignment across TMTs will be negatively related to EVP.

TMT age alignment

The extant literature has looked into the disparity in decision-making between younger and older executives (Taylor, 1975). Reasons cited for these differences include experience, motives and information processing capabilities. For example, Carpenter *et al.* (2004) note that younger or newer executives may rely more on their functional background, which might tend to dissipate over time due to exposure to other types of influences. In this vein, it has been found that average TMT age leads to differences in decision-making and overall firm strategy. For example, Tihanyi *et al.* (2000) find that average TMT age is negatively

associated with the global strategic posture of the firm. Furthermore, it has been noted that individuals from different age cohorts might have significantly different values and perspectives (Pfeffer, 1983). According to Hambrick *et al.* (1996), demographic differences could lead to dispersion in perspectives, which would make it less likely for individuals in different age cohorts to have the same model for firm strategy (Knight *et al.*, 1999).

In the environmental context, according to the theory of generativity, as people progress in age the desire for leaving a lasting legacy increases (McAdams and de St Aubin, 1992). Societal generativity that includes serving in roles of mentorship, providing leadership and generally contributing to the wellbeing of the subsequent generation increases with age (Snarey, 1993). Accordingly, do Paço *et al.* (2009) find a positive correlation between age and environmental sensitivity. It has been seen that environmental consumers frequently engaging in eco-friendly purchasing behavior tend to be older Roberts (1996). The author finds that older customers, as compared to younger customers, have a higher level of ecological awareness and a higher propensity to buy eco-friendly products. In line with generativity, the research base leveraging socio-emotional selectivity theory proposes an increase in the importance of social goals as opposed to materialistic gains in the context of self with aging (Kooij *et al.*, 2011). Increasing environmentally friendly behavior or selectively engaging in environmentally friendly behavior to advance social goals with age is an indicator of the propensity of sustainable decision-making in older TMTs (Kumar and Paraskevas, 2018; Kumar *et al.*, 2019). Given the differing levels of propensity for environmentally friendly actions by age, this paper theorizes that distinct TMTs at different levels of the age spectrum will struggle to evolve a common agenda on sustainable initiatives.

This paper proposes that the differences between younger and older executives will hamper behavioral integration between TMTs. UET states that executives make strategic choices through a process of gathering information, selecting a subset of information and interpreting that information (Hambrick and Mason, 1984). This paper theorizes that the process of selecting relevant information would vary between older and younger TMTs driven by differences in underlying motivations. For example, it is likely that older TMTs more interested in leaving a lasting legacy and less tied to their functional backgrounds (Carpenter *et al.*, 2004; McAdams and de St Aubin, 1992) will be selective in prioritizing information to further their goals of legacy creation. Accordingly, older TMTs might be more inclined to pursue environmental strategies that would negatively impact a firm's finances in support of their social goals. The differences in motivation would make the behavioral integration process more challenging as it depends on collaborative behavior and joint decision-making (Hambrick, 1994).

Effective communication between cohorts of different age groups could also become an issue for age misaligned buyer and supplier TMTs. As noted by Wiersema and Bantel (1992), according to organizational theorists there are costs associated with diversity of age within a group. Group members belonging to different age cohorts, for example, will find it more difficult to communicate as compared to same-cohort members. Such diversity in cohorts increases the likelihood of conflict and power struggles (Pfeffer, 1983). Higher levels of diversity in age, therefore, could lead to communications that are increasingly strained and conflict laden.

TMTs in similar age cohorts will likely share a similar propensity for environmental initiatives (McAdams and de St Aubin, 1992) and will likely experience superior behavioral integration. Similar aged cohorts would behave as complementary resources and according to the complementarity viewpoint, would be beneficial to the entire value chain. On the contrary, members of TMTs belonging to different age cohorts would lack the benefit of shared social goals compounded by issues related to communication styles leading to greater friction and conflict. Following the homogeneity arguments tied to complementarity that actions are most effective when buyer and supplier pairs act cohesively and collaboratively,

the age alignment across TMTs will lead to superior EVP while misalignment will be detrimental to overall EVP. Hypothesis three formally states the relationship between age alignment and EVP as follows:

H3. Higher levels of TMT age misalignment will be negatively related to EVP.

Buyer-supplier
TMT
misalignment
on EVP

Methods

Research setting

The empirical setting is publicly held U.S. manufacturing firms that are present in KLD's annual EVP ratings, the WRDS Compustat database and Bloomberg's SPLC database. The data collection began by extracting a complete listing of all manufacturing firms included in the Compustat database that had a 3-digit North American Industry Classification System (NAICS) code in the range from 311 to 339. Manufacturing firms in the oil and natural gas industry, the mining industry and the high-tech sector are also included, which requires inclusion of NAICS codes 211, 213, 518, 531 and 541. Manufacturing firms were considered ideal for our analysis as there is a clear environmental link between the output of a physical good produced by a supplier and sold to the buyer. In order to make a green product, both companies must work together to eliminate environmental waste throughout the supply chain. This contrasts with firms in more service-oriented industries such as banking or consulting which deal in less tangible goods and in which the dependence between a buyer and supplier for optimal environmental outcomes is less obvious. The resulting list of firms is then matched to the Bloomberg SPLC's database to yield a set of 550 manufacturing firms. Firms were then eliminated if they had missing or incomplete data in either the Bloomberg or the KLD or both databases. The final sample consists of 427 firms comprising 22 three-digit NAICS industries. The distribution of firms by NAICS is presented in Table 1. The analysis is performed at the level of the buyer-supplier dyad, meaning that each observation is a buyer-

NAICS code	Industry description	Total buyer-supplier dyads
211	Oil and Gas Extraction	83
213	Support Activities for Mining	29
311	Food Manufacturing	86
312	Beverage and Tobacco Product Manufacturing	17
313	Textile Mills	6
322	Paper Manufacturing	68
323	Printing and Related Support Activities	4
324	Petroleum and Coal Products Manufacturing	359
325	Chemical Manufacturing	891
326	Plastics and Rubber Products Manufacturing	3
327	Nonmetallic Mineral Product	12
331	Primary Metal Manufacturing	85
332	Fabricated Metal Product Manufacturing	81
333	Machinery Manufacturing	615
334	Computer and Electronic Product Manufacturing	3,033
335	Electrical Equipment, Appliance and Component Manufacturing	28
336	Transportation Equipment Manufacturing	1,949
337	Furniture and Related Product Manufacturing	14
339	Miscellaneous Manufacturing	80
518	Data Processing, Hosting and Related Services	18
531	Real Estate	22
541	Professional, Scientific and Technical Services	10

Table 1.
Summary of NAICS
industries used and the
number of
observations per
industry

supplier pair. In total there are 7,493 dyad-year observations for a seven-year period, 2009 through 2015. This is deemed as an acceptable period to conduct the analysis as significant levels of outsourcing were common during this time and it started after the 2008 financial crisis.

Dependent variable

The paper uses KLD ratings as the measure of a company's annual EVP profile. KLD data are used broadly for measuring not only EVP but also corporate social responsibility (CSR) and is widely regarded as the best available data for this purpose (Choi and Wang, 2009; Coombs and Gilley, 2005; Kacperczyk, 2009). KLD data include an evaluation of the firm's social performance in areas that include community, diversity, employment, environment and human rights (Servaes and Tamayo, 2013). Under each of these categories, KLD annually rates each company for the presence (or absence, hence a binary indicator) of several specific "strengths" (exemplary qualities) and for the presence (or absence) of specific "concerns" (problematic issues). For example, under the "environment" category, one of the strength items deals with clean energy and one of the concern items deals with environmental infractions. Empirical tests of the reliability and validity of the KLD annual ratings have been generally supportive (Sharfman, 1996; Walls *et al.*, 2012). The data are also used at length by institutional and individual social investors, further attesting to their perceived meaningfulness (Chin *et al.*, 2013).

To measure each firm's annual EVP, the environment category that gauge a firm's commitments to environmental issues were included. Under this category, KLD annually rates companies on a total of seven strengths and six concerns. The seven factors comprising strengths include beneficial product and services, recycling, pollution prevention, communications, clean energy, management systems and other strengths. KLD rates firms on six factors that comprise environmental concerns. These six factors comprising concerns include hazardous waste, ozone depleting chemicals, climate change, regulatory violations, agricultural chemicals and other concerns. The dependent variable is calculated as the net EVP score for each year: the sum of all the company's strengths minus all concerns under the environmental category. For instance, if a company in a given year had 3 strengths and 2 concerns, its EVP would be 1. The use of EVP net scores (KLD annual environmental strengths minus concerns) is consistent with the CSR net score used in Chin *et al.* (2013). The measure also supports the theoretical argument that TMT misalignment would affect both sides of the EVP metric. That is, TMT misalignment would affect investments in EVP strengths and/or risks in EVP weaknesses (concerns). The use of KLD net scores also follows the approach in extant studies (Choi and Wang, 2009; Deckop *et al.*, 2006; Wong *et al.*, 2011).

To measure each dyad's EVP, buyer firm's EVP is summed its supplier's EVP to get a unique yearly chain EVP for each buyer and supplier dyad. Bloomberg SPLC is used for supplier identification. Bloomberg maps about 35,000 companies with their suppliers and buyers by showing the flow of money between companies. The data provide linkages between a buyer and its suppliers, quantifying the percentage of a buyer's total cost of goods (COGs) sold being spent with each supplier and the percentage of each supplier's revenue attributable to the buyer's purchases. The data are collected from the Bloomberg terminals as a snapshot at the end of the year. Bloomberg collects supply chain information from various sources, including public filings [1], announcements from manufacturers and their suppliers and other proprietary data. Deriving information from a variety of data sources allows Bloomberg to capture supply chain data in detail. For example, for S&P 500 high-tech firms, the total number of suppliers identified by Bloomberg is on average seven times larger than that identified by Compustat. Not surprisingly, Bloomberg's supply chain data are starting to

get some attention in supply chain and operations research (Elking *et al.*, 2017; Kim and Davis, 2016; Schwieterman *et al.*, 2018; Schwieterman *et al.*, 2018; Steven *et al.*, 2014).

Independent variables

Top managers are identified through Bloomberg's executive profile database. The Bloomberg database includes both the current top managers and historical information on top managers of each firm. Bloomberg further includes career summaries for each executive listed in the database. These career summaries include the previous positions held by the executives as well as a brief description of every position.

This study draws support from the literature in calculating all the misalignment variables. Ndofor *et al.* (2015) for instance calculated their "competitive repertoire deviance" as the sum of squared difference in proportions of categories of competitive actions undertaken by the focal firm and the competitors. Similarly, Ferrier *et al.* (1999) calculated their "leader challenger action dissimilarity" as the sum of squared difference in proportions of categories of competitive action between the focal firm and the industry mean. This paper is looking at the buyer and supplier relationship from a dyadic perspective; hence, the dissimilarity measure is calculated as the sum of squared differences between comparative attributes of buyer firm and its suppliers. Dissimilarity is also normalized to account for the additional influence on buyer's decision-making from suppliers that supply a large percentage of the buyer firm's COGs sold. To account for the outsized voice of these large suppliers the dissimilarity measure is weighted by the percentage of the buying firm's COGs sourced through the specific supplier. Weighting is applied as the impact of a perfectly aligned dyad where the supplier is accountable for just one percent of the buyer's total spend may not have the same impact as a dyad with 50% of the buyer's supply volume. The data necessary for this weighting is available through Bloomberg SPLC.

The misalignment variables are scores of three different dissimilarity variables based on backgrounds of the TMT of the buyer firm and its suppliers. The first variable, functional dissimilarity, compares the TMT-member functional background of the buyer to those of its suppliers. To identify a manager's background for each firm and its suppliers the Bloomberg database's executive profiles is used for each manager in the TMT. Three major business operations functional backgrounds were sought: supply chain and operations management (SCOM), marketing and finance following Li *et al.* (2013). These three functional areas were chosen to give the analysis a broad measure of major business disciplines which also had enough representation on TMTs to give us a significantly large sample for the analysis. When a manager is found with any of these experiences, the variable is coded as one under each respective functional background, otherwise zero. For identifying executives under SCOM, the definition of Hendricks *et al.* (2014) is used. In order to quantify marketing background, managers were collected that at some point in their careers held positions in marketing, sales or advertising functions (Li *et al.*, 2013). For finance, identification is established through previous roles as financial officers as well as other positions related to finance such as, investment and corporate finance positions as defined in Li *et al.* (2013). This information is used to calculate the percentage of SCOM, marketing and finance related composition of each TMT, which is subsequently used to calculate the functional dissimilarity variable.

$$\text{Dissimilarity} - \text{Background} = (\text{Pct. Focal Firm COGS}) * [(x_{i,t} - x_{s,t})^2 + (y_{i,t} - y_{s,t})^2 + (z_{i,t} - z_{s,t})^2] \quad (1)$$

where i , s and t , denote focal firm, supplier firm and time (year) respectively, and x , y and z are the proportions of SCOM, marketing and finance experiences of buyer versus supplier TMT,

respectively. Eqn(2) provides a single score for each buyer and supplier dyad in a year. Zero values indicate perfect alignment with larger values indicating greater misalignment.

The second misalignment variable is measured in terms of gender dissimilarity of the buyer and supplier TMTs. To identify gender, Bloomberg's data on executive profiles are used for each member of the TMT. The percentage of males on each team is used to calculate gender dissimilarity:

$$\text{Dissimilarity} - \text{Gender} = (\text{Pct. Focal Firm COGS}) * (m_{i,t} - m_{s,t})^2 \quad (2)$$

where m is the percentage of males in each TMT.

The third misalignment variable is based on differences in age between TMTs. The Bloomberg database on executives is utilized to gather age information on all TMT members of all the firms in the sample. This variable, age dissimilarity, is calculated as follows:

$$\text{Dissimilarity} - \text{Age} = (\text{Pct. Focal Firm COGS}) * \left(\frac{a_{i,t} - a_{s,t}}{a_{i,t} + a_{s,t} / 2} \right)^2 \quad (3)$$

where a is the average age of the members of a TMT.

Control variables

A total of 13 control variables were included in the analyses along with the firm and year fixed-effects following the literature on TMT–EVP relationship (see Kumar and Paraskevas, 2018). Prior year's buyer EVP (lag *buyer EVP*) and the prior year's supplier EVP (lag *suppliers EVP*) were included to control for firm and supplier specific historical influences on EVP (Kumar *et al.*, 2019). Firm dependence (*power*) is included as a control since a power disadvantage in a relationship could put buyer firms at a disadvantage in their efforts to implement environmental initiatives throughout their supply chain. Following Elking *et al.* (2017), the variable is calculated as the difference between buyer firm's percentage of COGs expended on a supplier and the percentage of supplier's revenues sourced from the buyer. Power is then weighted by the supplier's importance to the buyer based on the percentage of the buying firm's COGs sourced through the specific supplier. Next *buyer size* is included by taking the log transformation of annual sales of a firm to correct for skew (Wowak *et al.*, 2015). Larger firms can access and invest more resources to increase their strengths and/or buffer against loss of strength ratings (Kumar *et al.*, 2019). Further, financially successful buyers may be more capable of investing in environmental initiatives. To control for this possibility, a profitability measure (*ROA*) is included as a control variable in the analysis. Following the same logic, *supplier size* and *supplier ROA* were also included as additional control variables. Bloomberg executive data are used to create three control variables, one for each of the functional backgrounds of interest in this study (SCOM, marketing and finance). These control variables account for a TMT that is dominated by one type of experience, as that may change environmental strategy. The gender composition of the TMT (*percentage male*) is also included as it has been suggested that females may view investments in environmental initiatives more positively in relation to their male counterparts (Brough *et al.*, 2016) and that gender of the TMT has EVP implications (Kumar and Paraskevas, 2018). The average age of the TMT (*TMT age*) is also included as a control variable. Lastly the size of the TMT is included and calculated as the count of number of executives (*TMT Size*) on the TMT. Firms with larger TMTs may have increasing difficulty in making decisions as more voices and viewpoints are added to the decision-making process.

Results

Summary statistics and pairwise correlations between variables are reported in Table 2. The estimated variance inflation factor (VIF) of the models suggests that the effects of

Variable	Mean	Std. Dev	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16
Chain EVP	233	2.40																
Lag buyer EVP (1)	1.07	1.94	1															
Lag supplier EVP (2)	0.77	1.35	0.144*	1														
Power (3)	1.06	5.78	0.042*	-0.0914	1													
Buyer size (4)	\$45435	\$55778	-0.11*	-0.011	0.148*	1												
Buyer ROA (5)	13.56	54.89	-0.02	-0.044*	0.021	0.043*	1											
Supplier size (6)	\$11097	\$27268	-0.048*	0.058*	-0.121*	0.021	0.005	1										
Supplier ROA (7)	14.01	107.99	-0.024*	0.008	-0.015	-0.005	0.008	0.015	1									
Percentage of TMT with finance Exp (8)	0.25	0.13	-0.153*	-0.024*	-0.023*	-0.146*	0.02	0.064*	-0.001	1								
Percentage of TMT with marketing Exp (9)	0.12	0.11	0.245*	0.075*	0.031*	0.012	-0.050*	-0.079*	0.005	-0.269*	1							
Percentage of TMT with SCOM Exp (10)	0.26	0.12	-0.091*	-0.031*	-0.056*	-0.100*	0.062*	-0.004	0.007	-0.034*	-0.175*	1						
Percentage of TMT that is male (11)	0.88	0.09	0.012	0.068*	-0.033*	-0.054*	-0.046*	0.011	0.017	0.087*	0.146*	-0.138*	1					
Average age of TMT (12)	53.17	3.23	0.074*	-0.036*	0.044*	0.115*	0.049*	0.039*	-0.025*	0.187*	-0.038*	0.029*	-0.066*	1				
Number of executives on the TMT (13)	14.24	5.68	0.074*	-0.090*	0.060*	0.271*	0.063*	0.009	-0.019	-0.334*	-0.086*	0.157*	-0.296*	0.052*	1			
Functional dissimilarity (14)	0.10	0.09	0.007	0.037*	-0.036*	-0.067*	-0.003	-0.061*	0.026*	0.140*	0.070*	0.050*	-0.025*	-0.031*	-0.179*	1		
Gender dissimilarity (15)	0.02	0.03	0.047*	-0.016	-0.013	-0.071*	0.024*	-0.045*	-0.012	0.075*	-0.111*	0.038*	-0.392*	0.056*	-0.061*	0.141*	1	
Age dissimilarity (16)	0.01	0.01	0.084*	-0.024*	-0.008	-0.050*	-0.033*	-0.059*	-0.017	-0.011	0.113*	-0.099*	0.072*	0.072*	-0.076*	0.02	-0.035*	1
Note(s): * $p < 0.05$																		

Table 2.
Means and correlations
between variable pairs

multi-collinearity are limited. Each model has a VIF estimate of less than 3 (2.84 the highest), which is deemed acceptable.

The regression analysis is done using an unbalanced panel dataset of 7,493 dyad-year observations comprised of 427 unique firms. To control for time and firm level effects, the models are run with fixed firm and year effects. Further, because of the possibility of industry heterogeneity, the estimated robust standard errors are clustered by industry NAICS codes. As indicated in the model specification, all appropriate variables are lagged by a year to control for contemporaneous correlations and endogeneity between the dependent and independent variables.

The regression results are given in [Table 3](#). Model 1 gives the results of only the control variables without any of the hypothesized relationships included. In Models 2 through 4, the dissimilarity variables are estimated individually together with the control variables. And finally, in Model 5, all three dissimilarity variables are estimated simultaneously.

Hypothesized relationships

The study hypothesizes that dissimilarity in functional background ([Hypothesis 1](#)), gender ([Hypothesis 2](#)) and age ([Hypothesis 3](#)) between buyer and its supplier's TMTs will have a negative effect on supply chain EVP. The functional dissimilarity variable has a negative and significant coefficient in both Models 2 and 5 ($-0.168, p < 0.01$; $-0.089, p < 0.01$, respectively). [Hypothesis 1](#) is therefore supported. The gender dissimilarity coefficient is negative and significant in both Models 3 and 5 ($-0.732, p < 0.01$; $-0.472, p < 0.01$). [Hypothesis 2](#) is therefore supported. The coefficients of age dissimilarity variable are negative in Models 4 and 5; however, they are not significant. [Hypothesis 3](#) is therefore not supported.

Control variables

Though not hypothesized, there are interesting results associated with the control variables. The coefficients for both the buyer and the supplier prior year's EVP is positive and significant in all Models. This suggests that firm specific historical influences on EVP affect future EVP. Higher percentage and the average age of female executives on the TMT have significant and positive coefficients on EVP. These are both in line with prior literature suggesting females, and older executives are more receptive to environmental initiatives ([Kumar and Paraskevas, 2018](#)). The analysis also finds that buyer firm power is positive and significant on EVP indicating that power residing with a buyer improves chain-level EVP. This result is particularly interesting and will be the subject of a post-hoc analysis in the discussion section.

Robustness and endogeneity checks

A concern with all empirical studies is the issue of endogeneity. Variables in a regression can be endogenous for several reasons including measurement errors, simultaneity/reverse causation and omitted variable bias ([Bascle, 2008](#); [Guide and Ketokivi, 2015](#); [Semadeni et al., 2014](#)). By lagging all independent variables, concerns associated with reverse causality are lessened, in that, by definition, the TMTs are already set before the EVP is assessed. To further reduce concerns of reverse causality, a commonly used estimation method; *two-stage residual inclusion* (2SRI) is performed ([Terza et al., 2008](#)). This method is performed in two-stages. First, regressions are used to predict the potentially endogenous variables; in this case functional dissimilarity, gender dissimilarity and age dissimilarity will all be predicted in the first stage. Next, the residuals (error term) from the first stage regression(s) are saved and entered into the second-stage regression as independent variables. Lastly the results are evaluated to ascertain if the originally hypothesized results hold ([Terza et al., 2008](#)). 2SRI is

Variables	Model 1 Chain EVP	Model 2 Chain EVP	Model 3 Chain EVP	Model 4 Chain EVP	Model 5 Chain EVP
Lag buyer EVP	0.328** (0.092)	0.329** (0.092)	0.328** (0.092)	0.328** (0.092)	0.329** (0.092)
Lag supplier EVP	0.095*** (0.005)	0.098*** (0.005)	0.098*** (0.005)	0.098*** (0.006)	0.100*** (0.006)
Power	0.002*** (0.001)	0.002*** (0.001)	0.002*** (0.001)	0.002*** (0.001)	0.002*** (0.001)
Buyer size	-0.289*** (0.045)	-0.292*** (0.046)	-0.289*** (0.044)	-0.291*** (0.046)	-0.292*** (0.045)
Buyer ROA	-3.06e-05 (0.0002)	-3.11e-05 (0.0002)	-2.41e-05 (0.0002)	-3.94e-05 (0.0002)	-3.07e-05 (0.0002)
Supplier size	3.38e-06 (2.22e-06)	3.39e-06 (2.22e-06)	3.38e-06 (2.22e-06)	3.40e-06 (2.22e-06)	3.40e-06 (2.21e-06)
Supplier ROA	0.0003*** (8.13e-05)	0.0003*** (8.18e-05)	0.0003*** (8.25e-05)	0.0003*** (8.28e-05)	0.0003*** (8.43e-05)
Percentage of TMT with finance Exp	2.534*** (0.381)	2.565*** (0.383)	2.527*** (0.375)	2.542*** (0.380)	2.550*** (0.378)
Percentage of TMT with marketing Exp	-1.077** (0.336)	-1.057** (0.330)	-1.073** (0.335)	-1.069** (0.331)	-1.061** (0.330)
Percentage of TMT with SCOM Exp	0.403 (0.402)	0.446 (0.404)	0.417 (0.405)	0.381 (0.406)	0.424 (0.423)
Percentage of TMT that is male	-3.361*** (0.327)	-3.382*** (0.325)	-3.451*** (0.318)	-3.352*** (0.304)	-3.425*** (0.310)
Average age of TMT	0.014* (0.007)	0.013 (0.007)	0.014* (0.007)	0.014* (0.007)	0.013 (0.007)
Number of executives on the TMT	0.0617*** (0.006)	0.0616*** (0.006)	0.0616*** (0.006)	0.0612*** (0.007)	0.061*** (0.007)
Functional dissimilarity		-0.168*** (0.031)			-0.089*** (0.015)
Gender dissimilarity			-0.732*** (0.066)		-0.472*** (0.090)
Age dissimilarity				-1.932 (1.202)	-0.896 (1.427)
Constant	3.010** (0.855)	3.106** (0.865)	3.117** (0.853)	3.068** (0.914)	3.157** (0.882)
Observations	7,493	7,493	7,493	7,493	7,493
R-squared	0.616	0.616	0.616	0.616	0.616
Note(s): Robust standard errors in parentheses *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$					

Table 3.
Regression results for
TMT dissimilarity and
supply chain EVP
performance
relationship

different than two-stage least squares (2SLS) in that instead of saving the fitted values from the first stage and adding them as an instrument in the second-stage regression, the residuals (error term) is saved from the first stage regression(s) and added to the second-stage regression as an independent variable.

The first stage regressions which estimate functional dissimilarity, gender dissimilarity and age dissimilarity (Table 4, Models 6, 7 and 8) use the independent variables used in the primary analysis in Table 3 minus the three dissimilarity measures. Two additional variables total inventory and gross profit are also added as there is an expectation that they will be significant predictors of the dissimilarity measures. Specifically, it is anticipated that firms with slack resources both from a product standpoint (inventory) and a financial standpoint (gross profit) may be less concerned with the potential impact of dissimilarity with supply chain partners and therefore may have higher levels of dissimilarity. In the second-stage regression (Table 4, Model 9) the residuals from the three first stage dissimilarity regressions are included as independent variables in re-estimating Model 5. Based on the results shown in Model 9, the coefficients for both functional dissimilarity and gender dissimilarity measures remain significant and negative. In other words, these results remain qualitatively similar as the results of the main models. This suggests that the estimates are consistent and unaffected by endogeneity (Davidson and MacKinnon, 1993). The regression outputs for this analysis can be found in Table 4.

Omitted variable bias can also be a source of endogeneity affecting the results. In order to address this concern, the impact threshold of a confounding variable (ITCV) is identified to validate the assertion of casual inference. Using the `konfound` command in Stata, it is determined that 67% of the functional dissimilarity – EVP relationship and 63% of the gender dissimilarity – EVP relationship would have to be as a result of bias in order to invalidate the results. Using the explanation by Frank (2000), 5,023 of the cases (observations) of the functional dissimilarity – EVP relationship would have to be replaced with an effect of 0 in order to invalidate the findings. Furthermore 4,704 of the cases in the gender dissimilarity – EVP relationship would have to be replaced with an effect of 0 in order to invalidate the findings. While exact thresholds for ITCV have not yet been established in the literature, we note that the result of the ITCV thresholds in this paper are equivalent to those found in previous research (Hill *et al.*, 2019) lending support to the conclusion that this analysis is not biased by omitted variables.

Another potential concern is that buyer-supplier pairs that have a similar orientation with both having strongly negative or positive views of the environment may not be as influenced by the dissimilarity between their TMTs. To address this concern, additional analysis is conducted by splitting the sample into subgroups. The first subgroup is comprised of buyer-supplier dyads where a low performing supplier is paired with a low performing buyer (Table 4, Model 11). Membership is made up of buyers that perform one SD below the mean for buyer EVP paired with its suppliers that perform one SD below the mean for supplier EVP. The second subgroup is comprised of high performing buyer and their supplier dyads with EVPs one SD above the respective mean EVPs (Table 4, Model 12). For the third subgroup, the remainder of dyads that are not members of the first or second subgroup are included comprising buyer-supplier dyads performing at the opposite end of the EVP spectrum (Table 4, Model 10). The regression results on the three subgroups is presented in Table 4. Based on the results, it can be concluded that the results are not being driven by EVP orientation alone and dissimilarity issues affect EVP performance.

Post-hoc analysis: the moderating influence of supply chain power. As a post-hoc analysis, the impact of buyer power is explored. There is a possibility that buyer power may overcome the coordination challenges as a result of misalignments between two supply chain members. Given the strategic nature of EM that requires participation across the supply chain, it is possible that buyer TMTs who are in a powerful position within the supply chain will be able

Variables	Model 6 Functional dissimilarity	Model 7 Gender dissimilarity	Model 8 Age dissimilarity	Model 9 Chain EVP	Model 10 Chain EVP	Model 11 Chain EVP	Model 12 Chain EVP
Lag buyer EVP	0.004*** (0.001)	0.001** (0.001)	0.001*** (0.24e-05)	0.320*** (0.090)	0.291** (0.087)	0.215* (0.052)	0.061 (0.065)
Lag supplier EVP	0.018*** (0.001)	0.005*** (0.001)	0.001*** (0.001)	0.124*** (0.003)	0.090*** (0.003)	0.235* (0.061)	0.028*** (0.006)
Power	-0.004*** (0.001)	-0.001** (0.001)	-0.001*** (1.22e-05)	0.001** (0.001)	0.002*** (0.001)	-0.025*** (0.003)	0.005*** (0.001)
Buyer size	-0.047*** (0.009)	-0.005 (0.006)	-0.003* (0.001)	-0.277*** (0.100)	-0.244*** (0.064)	0.697 (0.299)	1.598*** (0.377)
Buyer ROA	-9.01e-06 (3.98e-05)	1.01e-05 (8.02e-06)	-6.68e-06 (5.76e-06)	-0.001* (0.001)	0.001 (0.001)	-0.004 (0.002)	-0.012*** (0.003)
Supplier size	7.03e-08 (4.44e-08)	2.32e-09 (8.13e-09)	1.02e-08 (9.07e-09)	4.80e-06* (2.28e-06)	4.39e-06*** (6.69e-07)	-1.71e-06 (1.18e-06)	3.96e-06*** (9.61e-07)
Supplier ROA	-3.39e-06 (4.25e-06)	-2.12e-06 (2.08e-06)	1.07e-06*** (2.81e-07)	0.001** (8.23e-05)	0.001** (3.09e-05)	0.001* (9.15e-05)	0.001 (0.001)
Percentage of TMT with finance Exp	0.192** (0.071)	-0.013** (0.004)	0.003 (0.007)	2.570*** (0.308)	2.256*** (0.287)	11.84*** (1.069)	-0.344 (1.914)
Percentage of TMT with marketing Exp	0.158*** (0.029)	0.001 (0.004)	0.002 (0.003)	-1.158*** (0.289)	-1.595*** (0.408)	5.356 (3.007)	1.697 (2.394)
Percentage of TMT with SCOM Exp	0.264*** (0.068)	0.017* (0.008)	-0.009 (0.008)	0.475 (0.439)	0.320 (0.390)	-11.91** (1.314)	2.177* (0.754)
Percent of TMT that is male	-0.092 (0.116)	-0.123*** (0.017)	0.004 (0.013)	-3.464*** (0.286)	-2.692*** (0.274)	11.68 (14.06)	0.383 (0.471)
Average age of TMT	-0.007*** (0.001)	-0.001*** (0.001)	-0.001 (0.001)	0.007 (0.006)	0.013*** (0.004)	0.024 (0.118)	-0.105* (0.043)
Number of executives on the TMT	1.08e-05 (0.002)	-8.62e-05 (0.001)	-0.001 (0.001)	0.067*** (0.007)	0.044*** (0.010)	0.297* (0.071)	0.020 (0.027)
Total inventory	1.46e-06*** (1.92e-07)	1.49e-07*** (3.04e-08)	8.02e-08 (4.94e-08)				
Gross product	1.13e-06*** (1.34e-07)	1.79e-07 (1.38e-07)	8.25e-08*** (1.26e-08)				
Functional dissimilarity				-0.158*** (0.036)	-0.063*** (0.015)	3.045 (5.008)	-0.365 (0.207)
Gender dissimilarity				-0.505*** (0.113)	-0.589* (0.261)	21.09 (7.766)	-0.567*** (0.106)
Age dissimilarity				-1.041 (1.104)	-0.700 (1.506)	27.04 (33.66)	-3.357*** (0.378)
Functional dissimilarity error				-22.13*** (3.013)			
Gender dissimilarity error				38.34*** (6.671)			
Age dissimilarity error				-225.6*** (30.45)			
Constant	0.759*** (0.144)	0.196*** (0.032)	0.043 (0.031)	3.490** (1.345)	2.928*** (0.661)	-21.24 (13.27)	-3.237 (2.384)
Observations	7,806	7,795	7,735	7,493	7,180	98	215
R-squared	0.624	0.592	0.485	0.619	0.589	0.835	0.550

Note(s): Robust standard errors in parentheses
*** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$

Table 4.
Two-stage residual
inclusion and
robustness check

to wield that power to ameliorate the negative effect of TMT dissimilarity on EVP by forcing conformance. In order to test this, models 2 and 3 are replicated with the addition of an interaction term of functional dissimilarity * power and gender dissimilarity * power in their respective models.

In both cases there is a positive and significant interaction effect, indicating that buyer power is weakening the negative impact of dissimilarity on EVP. The interaction plots are included below to illustrate the findings.

Discussion

In this paper, the research objective was to explore linkages between misalignment in the TMT of two supply chain partners and EVP. The study finds that functional background misalignment inhibits EVP. In addition, the study also finds that gender misalignment between two supply chain partners serves as impediment to EVP. Although age misalignment was hypothesized to have a negative relationship with EVP, it was not supported. The significance of the findings on the impact of TMT misalignment and EVP helps explain some of the difficulties that firms face when leveraging suppliers to effectively achieve environmental goals. The research and practical implications are discussed next.

Our findings confirm prior studies embedded in UET and environmental studies suggesting that managerial functional experiences are important indicators of a firm's strategic choices, operational actions and environmental outcomes (for, e.g. [Hambrick and Wowak, 2019](#); [Kumar and Paraskevas, 2018](#); [Reimer et al., 2018](#); [Alexiev et al., 2010](#); [Chen et al., 2010](#); [Hambrick, 2007](#); [Hambrick et al., 1996](#)). More significantly, this finding extends the boundaries of UET in four different ways. Research in UET have determined that managerial functional experiences are important indicators of firm strategic and operational actions (for, e.g. [Hambrick, 2007](#); [Hambrick et al., 1996](#)). That is, managers conceive, interpret and act on information through their individual functional experiences. From its mathematical origin to applications investigating interfirm relationships, complementarity theory emphasizes asset alignment and complementation (for, e.g. [Paulraj et al., 2014](#); [Darnall and Edwards, 2006](#)). The functional background misalignment-EVP finding, which juxtaposes the two theories, suggests that complementarity can be built on the unique experiences of TMT members via complementary viewpoints engendering alignment of strategic goals and objectives. Second, research in supply chain management have determined that interests' alignment between supply chain partners are paramount for effective firm performance ([Eisenhardt, 1989](#); [Lambert, 2001](#); [Lambert and Cooper, 2000](#); [Lambert and Enz, 2017](#)). Indeed, alignment of priorities and objectives between a firm and its suppliers is considered the defining concept of supply chain management ([Selviaridis and Spring, 2018](#)). Our findings further suggest that interests are aligned when functional experiences align and thus consequently result in better EVP performance. Third, another stream of research in supply chain management suggests that when interests are misaligned, transactions costs (including agency costs) increases causing deviation from transaction cost's predicted mode of governance ([Handley, 2017](#)). This, research has established negatively impact performance. As our results are also outsourcing relationships, we can draw a parallel here. Our result suggests that when TMT experiences align, performance, specifically in the environmental space, improves. This implies aligning TMT experiences across chain members reduces transactions costs. And finally, academicians continue to examine how the network structure can help firms by providing them with unique advantages ([Dyer and Singh, 1998](#); [Zaheer and Bell, 2005](#)). Commonly acknowledged benefits include innovation, cost competitiveness, among others ([Dyer, 1996](#); [Jap, 1999](#)). This study contributes to this research stream by theorizing on how TMT alignment based on demographic homogeneity as an aspect of relational upside can help focal firms with their EVP.

In addition, the finding that EVP is associated with functional dissimilarity in this study is important to managers as it accentuates misalignment across firms as a key impediment to implementing the agenda of partner firms. In the worst case, a lack of alignment could sow discord through pursuit of divergent goals. As firms continue to expand their reliance on suppliers (Eppinger and Chitkara, 2006), they are becoming more dependent on them for achieving many of their goals. Simultaneously, issues of EM are of increasing importance (Lo, 2013; Rao and Holt, 2005; Tachizawa *et al.*, 2015; Zhu *et al.*, 2005) with EVP established as a source of competitive advantage (Kumar *et al.*, 2017). Given that TMT functional misalignment can be a barrier to leveraging suppliers to effectively achieve environmental goals, it is important for firms to explicitly account for TMT misalignment in strategic planning. TMTs play a key role in strategic direction for a firm, and therefore alignment between TMTs of firms working together becomes an important aspect of goal achievement.

Kumar and Paraskevas (2018) have highlighted some unique and differential focus related to EM initiatives based on TMT member's experience (operations, marketing, accounting and others) as well as age and gender characteristics. The preference for the type of EM initiative differs based on a member's background. Similar demographic representation across TMTs would lead to better integration of viewpoints and therefore lead to linked initiatives across the chain. In essence, behavioral integration from demographic similarity leading to shared priorities and strategic consensus would therefore help by creating linked initiatives that are similar and can be implemented across the chain as opposed to within a single entity. The benefits of strategic adoption of linked EM initiatives would impact EM outcomes across the supply chain beyond the boundaries of an individual firm. Findings supporting the upside of such linked initiatives have been emphasized by the positive impact of upstream and downstream integration by Vachon and Klassen (2006). It is therefore important that buyer-supplier TMTs have complementary views on an EM strategy and can benefit from relational demography to achieve objectives where similar initiatives undertaken with a unified purpose will result in gains that would be more than the sum of individual initiatives as emphasized by complementarity theory.

The general relationship between gender composition of TMTs and EVP with a focus on the negative relationship between gender misalignment of TMTs and EVP is interesting as well. First, the presence of female members on the TMT is found to have a strong positive relationship with EVP. This is a confirmation of prior studies that it is important to have female representation in TMTs to drive EVP (e.g. Kumar and Paraskevas, 2018). The results expand the boundaries of our understanding of the relationship between gender and EVP by suggesting that a) within firm gender diversification is good for EVP and b) across firms' gender misalignment is bad for EVP. In other words, the findings are suggesting that having a diversified gender on one TMT, and a homogenous gender (specifically without female representation) on another will inhibit the company's pursuit of EVP. Practically, along with complementary resources, need for raw materials, market access and several other dimensions examined in the extant literature as important factors in developing partnerships, the importance of TMT alignment should also be taken into account as part of planning and negotiations. If there is demographic misalignment between TMTs, then greater investments might be required for an appropriate communication strategy to avoid misunderstandings such as, more frequent strategic sessions between TMTs for a continuous evaluation of the joint strategy.

Finally, the *post-hoc* analysis of the power between a buyer and its supplier provide interesting insights. First, the results show that stronger firms are positioned to implement their environmental policies throughout the supply chain. Second and most importantly, it is found that buyer power has a weakening effect on the negative impact of dissimilarity on EVP. This finding is particularly interesting. Extant research has looked into the drivers of suppliers' development along the environmental dimension (Ehrgott *et al.*, 2013; Nath *et al.*, 2019).

Environmental initiatives are costly, and it is often the case that an unfair share of the environmental burden in a supply chain is shouldered by suppliers causing resentment (Kumar *et al.*, 2019). This finding contributes to this stream of literature suggesting that more powerful buyers often are able to achieve their EVP goals through pushing their agenda on the less powerful chain members. As shown in Figures 1 and 2, the power effect dominates both misalignment effects.

Figure 1.
Interaction plot–
functional dissimilarity
X power

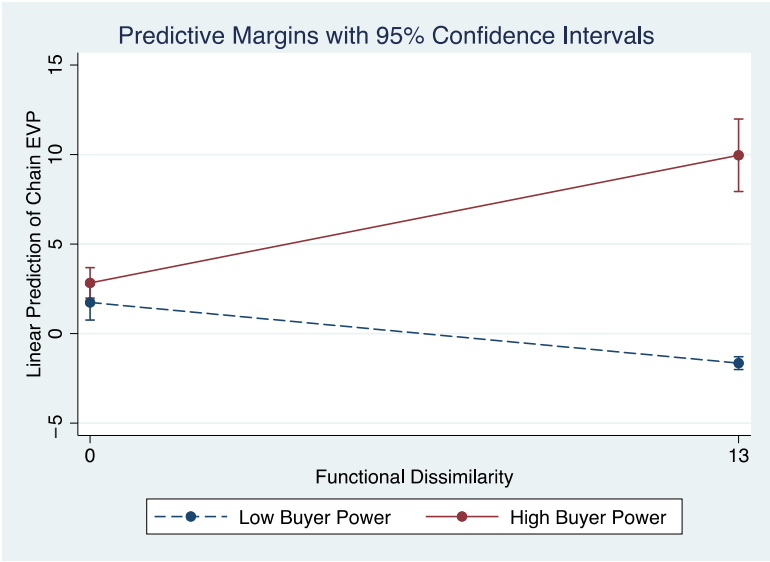
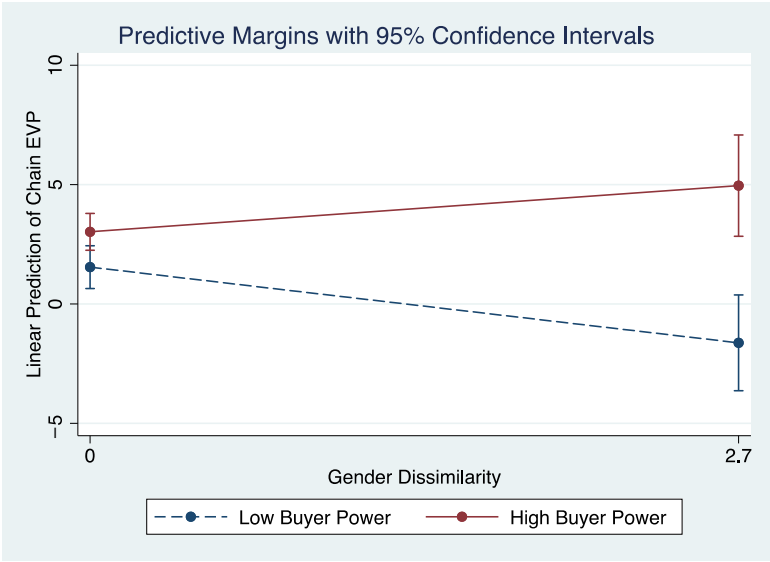


Figure 2.
Interaction plot–
gender dissimilarity
X power



In conclusion, the juxtapositions of UET and complementarity establish new lines of research: synergies of functional backgrounds of TMTs of supply chains. To the best of our knowledge, this is the first study that have looked beyond the TMTs of firms to comparisons of TMTs of the supply chain members.

Conclusions and limitations

Extant research has investigated the linkages between executive backgrounds and various performance metrics in the environmental space (Chin *et al.*, 2013; Petrenko *et al.*, 2016). Similarly, studies have highlighted the difficulties with supplier buy-in on environmental responsibilities (Foerstl *et al.*, 2015). However, little has been done, and no theoretical relationship has been established on the linkage between the (mis)match of buyer-supplier TMTs and EVP. In this research, by utilizing a panel dataset consisting of information on 427 unique firms from 2009 to 2015, this paper investigated the linkage between buyer-supplier TMT misalignment and supply chain EVP. This paper finds that misalignment in functional background and gender of TMTs negatively impacts EVP.

Although the findings provide unique insights into how supply chain TMT misalignment affects EVP, the research includes a few limitations. First, we have argued that misalignment between TMTs restricted by its members field of vision that is constrained by demographic characteristics negatively impact EVP. However, the study does directly measure the field of vision mechanism and a direct determination require further studies. Second, the study is limited to publicly traded firms due to data constraints. In general, public firms tend to be larger, tend to have more resources available to invest on green initiatives, are more powerful and are more often under the lens of stakeholders including regulators. Given these differences, suppliers could respond differently to initiatives from public firms. Thus, the impact on firm performance could differ for private and public firms. Third, the findings represent general patterns of the relationship between TMT misalignment and a firm's EVP, but these patterns may not apply unequivocally to every sector. Fourth, it must be acknowledged that the KLD data have certain limitation in terms of homogenizing the salience and significance of various environmental dimensions included in the rating, which might not be the case. This may have the potential of unwanted rationalization of EVP. These limitations are noted as future extensions of this research.

Note

1. Consequently, Compustat segment data are a subset of Bloomberg data.

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